

PRATHAM CHAUHAN

chauhanpratham812@gmail.com — +91 8837840132 — [LinkedIn URL](#) — [Portfolio Website](#)

EDUCATION

Thapar Institute of Engineering and Technology

B.E. Computer Science and Engineering – **Current CGPA: 7.56/10**

Patiala, Punjab

2022 – 2026

TECHNICAL SKILLS

- **Core Language:** C++ (Object Oriented Programming)
- **Project Technologies:** JavaScript (Node.js/Express), Python (FastAPI/Flask)
- **Tools & Environment:** Git, Docker (Basic Containers), Linux CLI, SQLite
- **Concepts:** API Routing, Basic Request Verification (HMAC), Data Deduplication

WORK EXPERIENCE

UrbanLock

Jan 2026 – June 2026

Founding Systems Engineer

- Built a MERN-stack backend platform with JWT authentication to securely handle locker bookings and digital wallet transactions.
- Created automated background tasks using Node-Cron jobs and event handlers to manage booking expirations, reminders, and real-time status updates.
- Integrated physical kiosk screens (DWIN displays) with the backend APIs over HTTP, ensuring smooth state synchronization even during poor network connectivity.
- Developed a central admin dashboard with data layouts to monitor revenue analytics and track the operational health of individual lockers.

Acmegrade

Aug 2023 – Oct 2023

Cybersecurity Summer Trainee

- Set up network traffic analysis scripts and simulated basic transport-layer attacks in a controlled lab environment to find application vulnerabilities.

PROJECTS

Event-Driven Notification Alert Bridge – *MacroDroid, Regex, Messaging API*

Jan 2026

- Built an event-driven automation framework utilizing regular expression text filters to ingest real-time local mobile system notifications and reliably forward verified transactions onto a secondary network.
- Maintained an entirely local execution flow to guarantee offline data containment, minimizing structural data leakage fields while successfully replacing legacy commercial sound hardware endpoints.

Secure Webhook Ingestion API – *Python, FastAPI, Docker, SQLite*

Dec 2025 – Jan 2026

- **API Security & Validation:** Verified webhook payloads using HMAC-SHA256 signatures with constant-time verification to prevent timing attacks, using Pydantic schemas to strictly validate data structures.
- **Idempotency Engine:** Enforced database-level primary key constraints to deduplicate incoming requests, ensuring safe at-least-once message processing without data corruption under heavy loads.
- **Observability & Containers:** Added custom middleware to track request latencies and core application metrics, containerizing the application with Docker to ensure clean deployment.

ChatMinds: Deterministic NLP Engine – *Python, Flask, HTML, CSS, JSON*

Jan 2024 – May 2024

- **Hybrid Pipeline & Routing:** Designed a query-routing system that combines simple regular-expression matching for common intents with a fallback TF-IDF vector index for semantic search.
- **Performance Optimization:** Setup simple text filters to clear out non-critical greeting words at startup, reducing unnecessary processing overhead and routing technical questions efficiently.
- **Client Protection:** Mitigated common Cross-Site Scripting (XSS) risks by enforcing sanitization checks on input fields before the user data reached the backend.

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** AWS Academy Cloud Security Foundations & AWS Academy Cloud Builder.
- **Achievements:** Bronze Medalist, District-level Karate.
- **Volunteering:** Class In-charge at Pratigya Abhiyan; managed operational resource coordination for over 70 students.